

Introductory Slides

Consciousness In Language Models, Uncertainty-aware
Language Models, Computer Vision

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Profile

Mohit Joshi

Topics: Consciousness in Language Models, Non-Rewardable Reinforcement Learning, Uncertainty-aware Language Models, Computer Vision

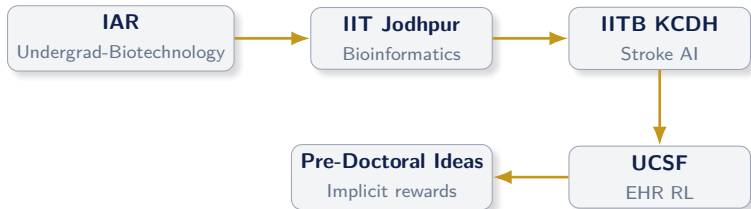
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Undergrad (Biotechnology Biophysics) → Bioinformatics → stroke segmentation → sequential EHR treatment modeling → non-rewardable RL

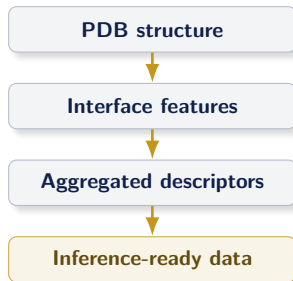
Research Trajectory



IIT Jodhpur

Protein-DNA affinity interface
analysis pipeline

- Automated multichain PDB processing.
- Combined Python, Fortran, shell tools, and structural biology utilities.
- Built the computational foundation for later ML work in medical and biological data.



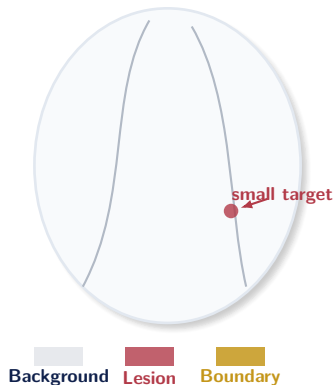
KCDH

Problem

Small ischemic lesions can occupy a tiny fraction of the image volume.

Voxel-overlap losses can remain numerically acceptable while clinically important small lesions are missed entirely.

Work done during exchange semester and research internship at IIT Bombay KCDH.



Size Penalty Loss: The Idea

Size-Conditioned Objective

$$\mathcal{L}_{size} = \exp\left(-\frac{V_g}{\tau_s}\right) \cdot \frac{|V_p - V_g|}{V_g + \epsilon}$$

Soft volume

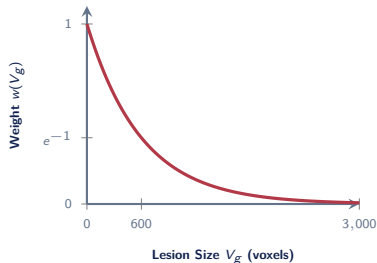
$$V_p = \sum p_i$$

True volume

$$V_g = \sum y_i$$

Weighting

Scales up for minute targets.



Dice / Tversky / Focal

+ Size Penalty ($\mathcal{L}_{size}$)

Joint Training Objective

The loss adds a size-conditioned volume signal

From Overlap Losses To Size-Aware Regularization

**Voxel Overlap
(Local Agreement)**



**Minute Lesions
(Often Missed)**



**Volume-Aware
Regularization**

What stays the same

3D segmentation backbone
and standard spatial losses.

What changes

An auxiliary term that explicitly
reacts to lesion size.

Why it matters

Minute-target segmentation
needs size fidelity, not just
overlap.

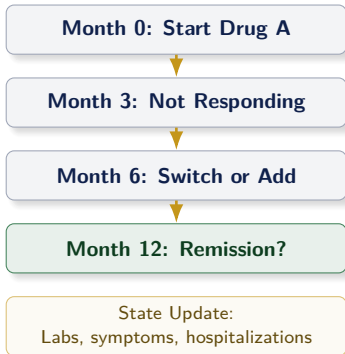
UCSF

Central question

Drug repurposing is often framed as a static, single treatment decision. The clinical story is actually **sequential**: treatment is revised until remission or failure.

- **Standard emulation:** drug A vs drug B at one time step.
- **Sequential view:** continue, switch, add, or escalate over time.

Each decision should react to the updated patient state.



From T+1 Emulation To Sequential Policies

Standard setup

Emulate one treatment contrast at a single time step

Research direction

Emulate a sequence of treatment decisions over evolving states

Baseline State $\rightarrow$ Drug A vs B

State S_t



Drug Action A_t



Next State S_{t+1}

The deeper question is whether drug repurposing should be modeled as a static single decision or as a dynamic treatment trajectory.

Implicit Non-Rewardable RL

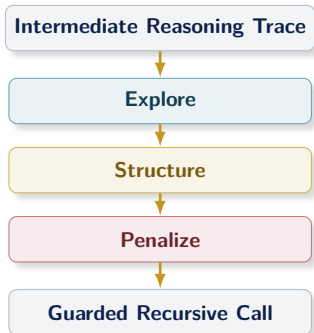
Research question

Can useful guardrails for **Recursive Language Models** be learned from implicit rewards in reasoning trajectories, rather than hard-coded constraints?

Recursive LMs

Implicit Rewards

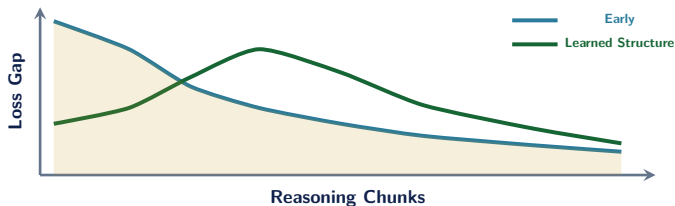
Epiplexity



Epiplexity As A Measurement Axis

Epiplexity Formulation

$$\text{Epiplexity} \approx \sum_i (\mathcal{L}_{\text{early}}(x_i) - \mathcal{L}_{\text{final}}(x_i))$$



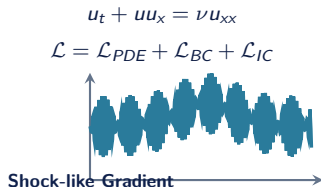
High Risk High Reward: we attempt to identify behavior-level reward proxies, test whether they transfer across domains, and apply them as guardrails for recursive reasoning.

Serendipity

Serendipity: Scientific Computing

PINNs

Physics-Informed Neural Networks for PDEs



PySR and SINDy

Equation recovery for epidemiology SIR dynamics

$$\begin{aligned}\dot{S} &= -\beta SI, \\ \dot{I} &= \beta SI - \gamma I, \\ \dot{R} &= \gamma I.\end{aligned}$$



These projects keep mathematical biology and scientific computing as active exploratory directions.

Find Me At!

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